

Random Forest for classification:

Output:

Class	Precision	Recall	F1-Score	Support
0	0.92	0.91	0.92	79
1	0.83	0.85	0.84	41
Accuracy	-	-	0.89	120
Macro Avg	0.88	0.88	0.88	120
Weighted Avg	0.89	0.89	0.89	120

- 1) what is the accuracy value for this **random forest**? - **0.89**
- 2) what is the correct classification of **non_purchase** total input **non_purchase** in test set? - **0.91 (Recall)**
- 3) what is the correct classification of **purchase** total input **purchase** in test set? – **0.85 (Recall)**
- 4) what is the percentage of correct classify of **non_purchase** to the sum of correct & wrong classifications **non_purchase** in testset? – **0.92 (Precision)**
- 5) what is the percentage of correct classify of **purchase** to the sum of correct & wrong classifications **purchase** in testset? – **0.83 (Precision)**
- 6) what is the overall performance of **non_purchase** ? - **0.92 (F1 Measure)**
- 7) what is the overall performance of **purchase** ? - **0.84 (F1 Measure)**
- 8) what is the average performance of **Precision**(correctly & wrongly classified)? – **0.88(Macro Average value for precision)**
- 9) what is the average performance of **Recall**(correctly classified)? – **0.88 (Macro Average value for Recall)**
- 10) what is the average performance of **F1-Measure (overall performance)**? – **0.88 (Macro Average value for F1-Measure)**
- 11) What is the sum of product of proportion rate(weight) of each class **Weighted Average**?

Precision – 0.89

Recall – 0.89

F1-Measure – 0.89

Decision Tree for classification:

Output:

Class	Precision	Recall	F1-Score	Support
0	0.96	0.90	0.93	79
1	0.83	0.93	0.87	41
Accuracy	-	-	0.91	120
Macro Avg	0.89	0.91	0.90	120
Weighted Avg	0.91	0.91	0.91	120

- 1) what is the accuracy value for this **Decision Tree**? - **0.91**
- 2) what is the correct classification of **non_purchase** total input **non_purchase** in test set? - **0.90 (Recall)**
- 3) what is the correct classification of **purchase** total input **purchase** in test set? – **0.93 (Recall)**
- 4) what is the percentage of correct classify of **non_purchase** to the sum of correct & wrong classifications **non_purchase** in testset? – **0.96 (Precision)**
- 5) what is the percentage of correct classify of **purchase** to the sum of correct & wrong classifications **purchase** in testset? – **0.83 (Precision)**
- 6) what is the overall performance of **non_purchase** ? - **0.93 (F1 Measure)**
- 7) what is the overall performance of **purchase** ? - **0.87 (F1 Measure)**
- 8) what is the average performance of **Precision**(correctly & wrongly classified)? – **0.89(Macro Average value for precision)**
- 9) what is the average performance of **Recall**(correctly classified)? – **0.90 (Macro Average value for Recall)**
- 10) what is the average performance of **F1-Measure** (overall performance)? – **0.91 (Macro Average value for F1-Measure)**
- 11) What is the sum of product of proportion rate(weight) of each class **Weighted Average**?

Precision – 0.91

Recall – 0.91

F1-Measure – 0.91

SVM for classification:

Output:

Class	Precision	Recall	F1-Score	Support
0	0.77	0.97	0.86	79
1	0.90	0.44	0.59	41
Accuracy	-	-	0.79	120
Macro Avg	0.83	0.71	0.73	120
Weighted Avg	0.81	0.79	0.77	120

- 1) what is the accuracy value for this **SVMC**? - **0.79**
- 2) what is the correct classification of **non_purchase** total input **non_purchase** in test set? - **0.97 (Recall)**
- 3) what is the correct classification of **purchase** total input **purchase** in test set? – **0.44 (Recall)**
- 4) what is the percentage of correct classify of **non_purchase** to the sum of correct & wrong classifications **non_purchase** in testset? – **0.77 (Precision)**
- 5) what is the percentage of correct classify of **purchase** to the sum of correct & wrong classifications **purchase** in testset? – **0.90 (Precision)**
- 6) what is the overall performance of **non_purchase** ? - **0.86 (F1 Measure)**
- 7) what is the overall performance of **purchase** ? - **0.59 (F1 Measure)**
- 8) what is the average performance of **Precision**(correctly & wrongly classified)? – **0.83 (Macro Average value for precision)**
- 9) what is the average performance of **Recall**(correctly classified)? – **0.71 (Macro Average value for Recall)**
- 10) what is the average performance of **F1-Measure** (overall performance)? – **0.73 (Macro Average value for F1-Measure)**
- 11) What is the sum of product of proportion rate(weight) of each class **Weighted Average**?

Precision – 0.81

Recall – 0.79

F1-Measure – 0.77

Finally, **Decision tree accuracy** is a **good prediction** and value is : **0.91**